

Question:

Find the sum of all positive integers n such that $n+2$ divides the product $3(n+3)(n^2+9)$.

Reasoning + Answer:

<think>

1. **Understand the given problem**: We need to find the sum of all positive integers n such that $n+2$ divides the product $3(n+3)(n^2+9)$.
2. **Set up the divisibility condition**: Since $n+2$ divides $3(n+3)(n^2+9)$, we can write:
$$3(n+3)(n^2+9) = k(n+2)$$
for some integer k .
3. **Analyze the divisibility**: For $n+2$ to divide $3(n+3)(n^2+9)$, it must also divide the product modulo $n+2$. We will use modular arithmetic to simplify the problem.
4. **Consider the polynomial** $n+2 \equiv 0 \pmod{n+2}$: This implies $n \equiv -2 \pmod{n+2}$.
5. **Substitute** $n \equiv -2 \pmod{n+2}$: Replace n with -2 in the polynomial $3(n+3)(n^2+9)$:
$$3((-2)+3)((-2)^2+9) \pmod{n+2}$$
6. **Simplify the expression**:
$$3(1)(4+9) \pmod{n+2}$$
$$3 \cdot 13 \pmod{n+2}$$
$$39 \pmod{n+2}$$
7. **Determine when** $n+2$ divides 39 : We need $n+2$ to divide 39 . The divisors of 39 are $\pm 1, \pm 3, \pm 13, \pm 39$.
8. **Find the corresponding** n :
$$n+2 = 1 \implies n = -1 \quad (\text{not positive})$$
$$n+2 = 3 \implies n = 1 \quad (\text{positive})$$
$$n+2 = 13 \implies n = 11 \quad (\text{positive})$$
$$n+2 = 39 \implies n = 37 \quad (\text{positive})$$
9. **Sum the valid positive** n :
$$n = 1, 11, 37$$
$$\text{Sum} = 1 + 11 + 37 = 49$$
10. **Conclusion**: The sum of all positive integers n such that $n+2$ divides $3(n+3)(n^2+9)$ is 49 .

<answer> 49 </answer>